# IGCSE 0607 Practice Questions — E7.6

## Pythagoras' theorem and trigonometry in 3D

**Total Questions:** 3 — **Total Marks:** 19 Marks

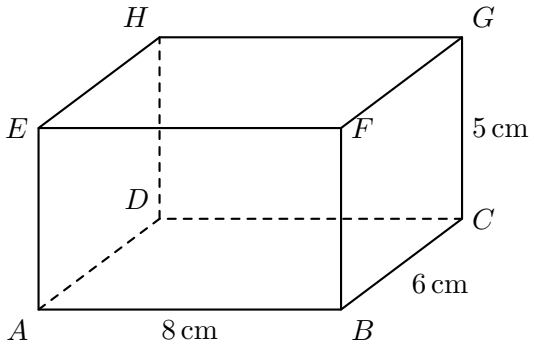
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**1** The diagram shows a cuboid $ABCDEFGH$.
$AB = 8$ cm, $BC = 6$ cm and $CG = 5$ cm.




**(a)** Calculate the length of the diagonal $AC$.

..... cm **[2]**

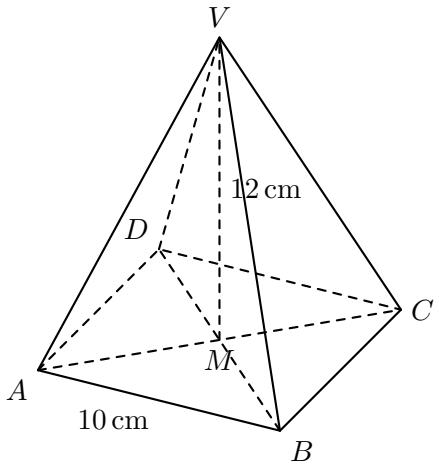
**(b)** Calculate the length of the space diagonal $AG$.

..... cm **[2]**

**(c)** Calculate the angle between the line $AG$ and the base $ABCD$.

..... ° **[2]**

**2** The diagram shows a pyramid $VABCD$ with a square base $ABCD$ of side 10 cm.
The point $M$ is the centre of the base and $V$ is vertically above $M$.
The vertical height $VM$ of the pyramid is 12 cm.



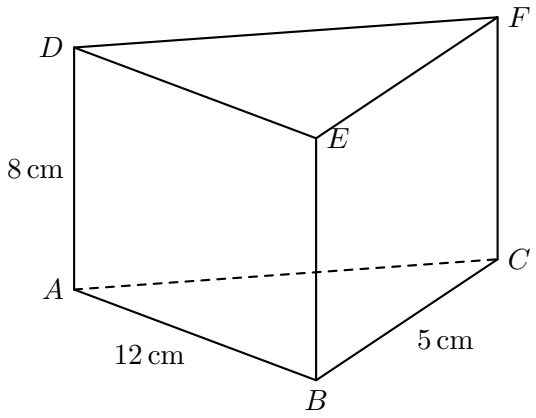

**(a)** Calculate the length of the edge $VA$.

..... cm **[3]**

**(b)** Calculate the angle between the edge $VA$ and the base $ABCD$.

..... ° **[2]**

**3** The diagram shows a right triangular prism $ABCDEF$.
The base $ABC$ is a horizontal right-angled triangle with angle $ABC = 90°$.
$AB = 12$ cm, $BC = 5$ cm and the vertical height $AD = 8$ cm.




**(a)** Calculate the length of $AF$.

..... cm **[3]**

**(b)** Calculate the angle between the line $AF$ and the base $ABC$.

..... ° **[2]**

**(c)** Calculate the angle between the line $AF$ and the vertical plane $ABED$.

..... ° **[3]**